

6CCS3PRJ

Designing for Consent Revocation in Online Platforms: WhatsApp Photo

Sharing

Final Project Report — Appendices

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About this document. This file contains the appendices that accompany the main project report (*report.pdf*). It is submitted as a separate PDF in line with the university's submission instructions. Cross-references of the form "Section X" or "Chapter X" refer to the main report unless stated otherwise.

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Appendix A

Thematic Analysis Materials

This appendix reproduces the reflexive thematic analysis workbook as exported from the project Miro board.

A.1 Interview extracts board (raw)

First page of the full colour-coded transcript extract board.



A.2 Inductive codebook (initial iteration)

First page of the initial 56-code codebook produced through close reading of the interview transcripts.


Control and Agency
Understandibility and Clarity
Trust and Transparency
Social Emotional Impact
Forwarding/Redistribution
Co-Subject Multi-Party Privacy
UI Design Feedback

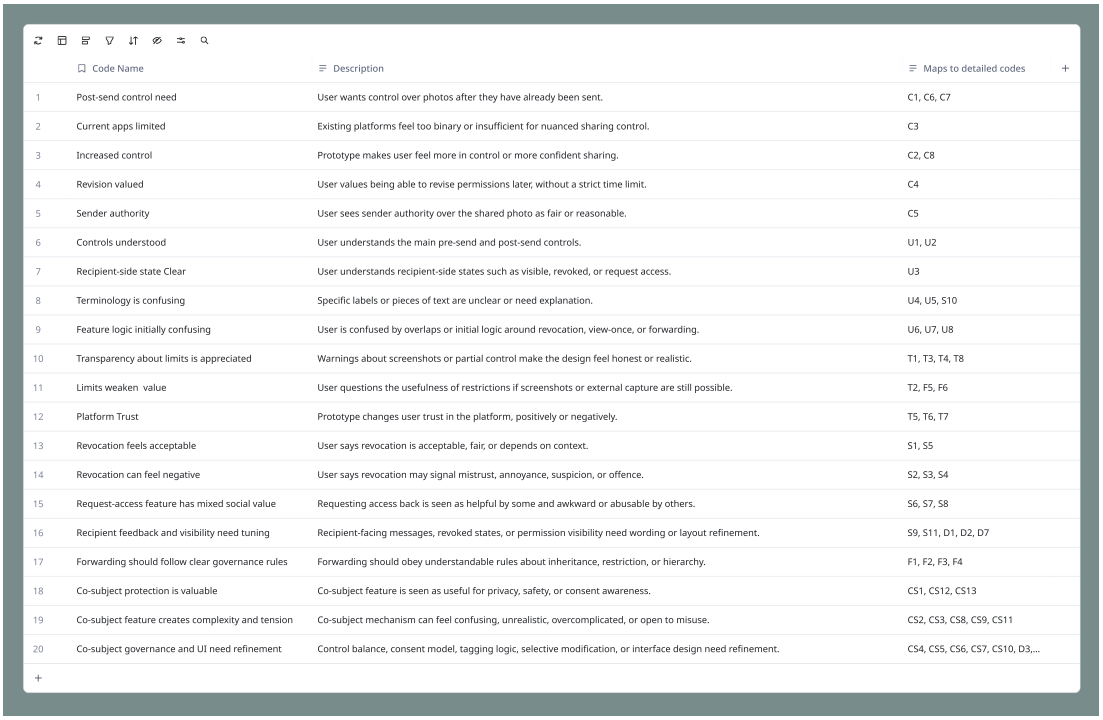
A.3 Interview extracts (refined subset)

First page of the refined extract set used for focused coding.



A.4 Working codebook (simplified)

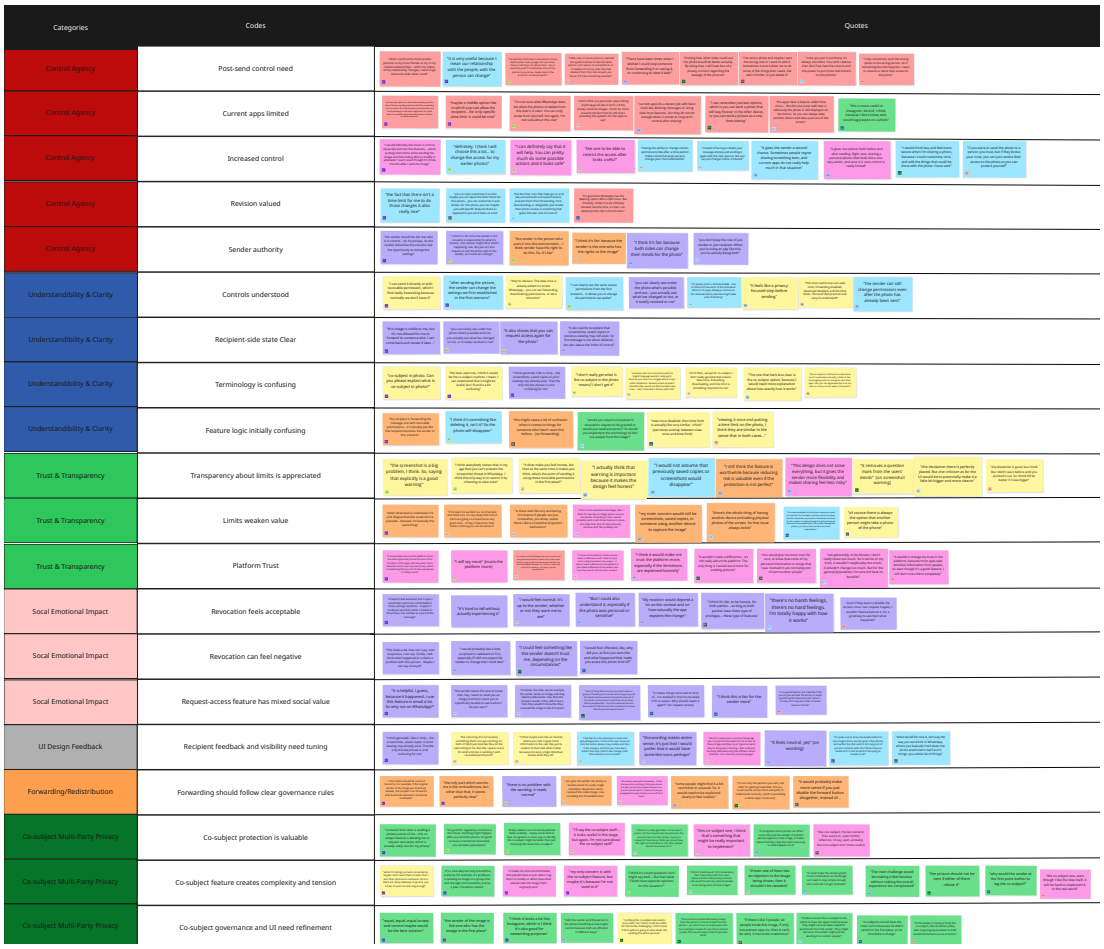
First page of the simplified 20-code working codebook used for cross-case tagging.



	Code Name	Description	Maps to detailed codes
1	Post-send control need	User wants control over photos after they have already been sent.	C1, C6, C7
2	Current apps limited	Existing platforms feel too binary or insufficient for nuanced sharing control.	C3
3	Increased control	Prototype makes user feel more in control or more confident sharing.	C2, C8
4	Revision valued	User values being able to revise permissions later, without a strict time limit.	C4
5	Sender authority	User sees sender authority over the shared photo as fair or reasonable.	C5
6	Controls understood	User understands the main pre-send and post-send controls.	U1, U2
7	Recipient-side state Clear	User understands recipient-side states such as visible, revoked, or request access.	U3
8	Terminology is confusing	Specific labels or pieces of text are unclear or need explanation.	U4, U5, S10
9	Feature logic initially confusing	User is confused by overlaps or initial logic around revocation, view-once, or forwarding.	U6, U7, U8
10	Transparency about limits is appreciated	Warnings about screenshots or partial control make the design feel honest or realistic.	T1, T3, T4, T8
11	Limits weaken value	User questions the usefulness of restrictions if screenshots or external capture are still possible.	T2, F5, F6
12	Platform Trust	Prototype changes user trust in the platform, positively or negatively.	T5, T6, T7
13	Revocation feels acceptable	User says revocation is acceptable, fair, or depends on context.	S1, S5
14	Revocation can feel negative	User says revocation may signal mistrust, annoyance, suspicion, or offence.	S2, S3, S4
15	Request-access feature has mixed social value	Requesting access back is seen as helpful by some and awkward or abusable by others.	S6, S7, S8
16	Recipient feedback and visibility need tuning	Recipient-facing messages, revoked states, or permission visibility need wording or layout refinement.	S9, S11, D1, D2, D7
17	Forwarding should follow clear governance rules	Forwarding should obey understandable rules about inheritance, restriction, or hierarchy.	F1, F2, F3, F4
18	Co-subject protection is valuable	Co-subject feature is seen as useful for privacy, safety, or consent awareness.	CS1, CS12, CS13
19	Co-subject feature creates complexity and tension	Co-subject mechanism can feel confusing, unrealistic, overcomplicated, or open to misuse.	CS2, CS3, CS8, CS9, CS11
20	Co-subject governance and UI need refinement	Control balance, consent model, tagging logic, selective modification, or interface design need refinement.	CS4, CS5, CS6, CS7, CS10, D3,...

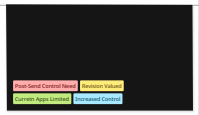
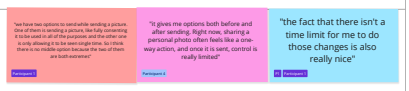
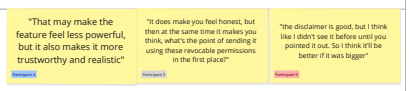
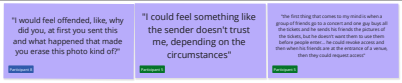
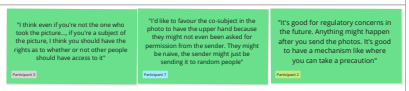
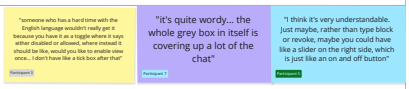
A.5 Quote-to-code coding sheet

First page of the quote-to-code mapping sheet.



A.6 Final themes table

First page of the consolidated themes table (definitions, codes, and example quotations).

Theme	Description	Codes	Example Interview Quotes
Theme 1: From Binary to Revocable Consent	This theme concerns participants' views on current photo-sharing controls as too limited or too extreme. It captures their preference for permission settings that can be adjusted both before and after sending, rather than fixed at the moment of disclosure.		
Theme 2: Bounded Control and Honest Limitations	This theme concerns how participants responded to the system's communication of its own limits, particularly around screenshots and saved copies. It captures the tension between valuing transparency and questioning the usefulness of controls that cannot offer complete protection.		
Theme 3: Revocation as Social Negotiation	This theme concerns how participants interpreted photo revocation as an interpersonal action rather than a purely technical one. It includes responses shaped by trust, fairness, awkwardness, and views on the request-access mechanism.		
Theme 4: Co-Subject Privacy as a Multi-Consent Problem	This theme concerns participants' reactions to giving depicted individuals a role in post-send control. It includes both support for the underlying privacy principle and concerns about feasibility, power balance, and how such control should be governed.		
Theme 5: Interface Legibility and Permission Understandibility	This theme concerns how clearly the interface communicated permission states, feature logic, and available actions. It includes participant responses to wording, repetition, layout, and the balance between visibility and message-space efficiency.		

Appendix B

Interview Guide

This appendix records the scenario-based moderator script used in the evaluation interviews. The wording follows the version spoken in sessions (including the consent statement and scenario prompts). Each numbered question lists the **linked research questions** (Main RQ and RQ1–RQ5) taken from the analysis mapping used alongside the guide: RQ1 addresses current platform limitations; RQ2 expectations and post-sharing concerns; RQ3 clarity of the interface and permission model; RQ4 fairness and social implications; RQ5 trust, safety, control, and realism of revocation. The main research question is stated in Chapter 1.

Research question key

Main RQ

Overall contribution of the proposed design.

RQ1

Limitations of current platforms for controlling shared photos.

RQ2

User expectations, practices, and concerns about post-sharing control.

RQ3

Clarity and understandability of the interface and permission model.

RQ4

Social implications for stakeholders.

RQ5

Trust, safety, perceived control, and realism of revocation.

B.1 Section 0: Mandatory consent statement

Say verbatim:

> "Before we begin, I'd like to confirm your consent to participate in this interview. This interview is part of my research on how messaging platforms might support consent revocation for shared photos. The purpose of this interview is to understand your views and reactions to a design concept. During the interview, I will show you example screens from a prototype. I am not testing you — I am testing the design. There are no right or wrong answers, and honest reactions are very helpful. Some questions are about privacy, control, and what happens to photos after they are shared. You can skip any question or stop the interview at any time. With your permission, I would like to audio-record this interview so I can analyse the responses later. The recording will only be used for this project, and your responses will be anonymised. You can choose to stop the recording or withdraw from the interview at any time. If you're happy to proceed, could you please state your full name and age, and confirm that you agree to be interviewed and recorded?"

B.2 Section 1: Warm-up and current photo-sharing behaviour

1. **Can you tell me a bit about how you usually share photos on apps like WhatsApp?**

   Linked: RQ2.

   Probes optional: private vs group chats; casual vs personal photos; frequency.

2. **When you send a photo to someone, do you usually think about what might happen to it afterwards? If yes, what kinds of concerns or challenges come to mind?**

   Linked: RQ2.

   Probes optional: saving, forwarding, screenshots, long-term retention; active vs incidental

thought.

3. **Have you ever wanted more control over a photo after you had already sent it?**

   Linked: RQ1; RQ2.

   Probes optional: type of control wished for; whether the app helped.

4. **What do you think current apps do well or badly when it comes to controlling shared photos?**

   Linked: RQ1.

B.3 Section 2: Scenario 1 — Before sending a photo

Show: photo composer / permission setup (view-once, forwarding, download, co-subject, time limit; note on screenshots and saves).

5. **What do you think is happening on this screen?**

   Linked: RQ3.

   Probes optional: purpose of controls; what the sender decides.

6. **What do you think each of these settings means?**

   Linked: RQ3.

   Probes optional: view once; forwarding disabled; download disabled; time limit; co-subject in photo.

7. **Which of these options feel useful to you, and which feel unnecessary or unclear?**

   Linked: RQ2; RQ3.

   Probes optional: realistic pre-send use; effort; missing options.

8. **There is also a warning here saying that screenshots and saves cannot be undone. How do you feel about that?**

   Linked: RQ3; RQ5.

   Probes optional: honesty vs usefulness; whether the warning is wanted.

Moderator note: Only explain controls if the participant is completely stuck; prior interpretation is the primary data.

B.4 Section 3: Scenario 2 — Changing permissions after sending

Show: sent photo with photo menu (revoke/restore, forwarding, download, time limit).

9. **Looking at this screen, what do you think the sender is able to do now?**
   Linked: RQ3.
   Probes optional: what is changeable after send; comparison to deleting a message; revoke vs permission change.

10. **What would you expect to happen if the sender revokes access here?**
    Linked: RQ3; RQ5.
    Probes optional: visibility; recipient notification; offline copies; if already seen.

11. **How understandable does this post-send control feel to you?**
    Linked: RQ3.
    Probes optional: clarity of change; menu affordances; discoverability.

12. **Would this kind of feature make you feel more in control after sending a photo? Why or why not?**
    Linked: RQ5.
    Probes optional: trust; anxiety; realistic use.

Moderator note: Keep focus on clarity, expectations, realism, and perceived control rather than implementation detail.

B.5 Section 4: Scenario 3 — Recipient view, request access, and fairness

Show: recipient-side revoked placeholder (screenshots / prior viewing note).

13. **What do you think this screen is telling the recipient?**
    Linked: RQ3; RQ4.
    Probes optional: clarity of change; sufficiency of explanation.

14. **How would you feel if you were the recipient in this situation?**
    Linked: RQ4; RQ5.

Probes optional: reasonable vs awkward; fairness tone.

15. **What do you think about the idea that the recipient can request access again?**

    Linked: RQ4.

    Probes optional: comfort as sender or recipient; balance.

16. **Does this feel fair to both people, or more fair to one side than the other?**

    Linked: RQ4.

    Probes optional: sender power; explanation; objection rights.

17. **What do you think about the wording used here?**

    Linked: RQ4.

    Probes optional: neutrality vs blame.

Moderator note: Do not defend the revoked state; negative reactions are informative.

B.6 Section 5: Scenario 4 — Forwarding and residual concerns

Show: forwarding behaviour and secondary recipient where relevant; request-access flow and sender accept/reject if shown.

18. **What do you think this forwarding feature is trying to do?**

    Linked: RQ3.

    Probes optional: blocked vs allowed forwarding; sender rules after forward.

19. **How do you feel about the idea that the sender's rules continue even after the photo is forwarded?**

    Linked: RQ3; RQ5.

    Probes optional: usefulness; realism; over-control.

20. **Even with these features, what concerns would still remain for you?**

    Linked: RQ1; RQ2; RQ5.

    Probes optional: screenshots; prior downloads; residual value of imperfect control.

B.7 Section 6: Scenario 5 — Co-subject request for revocation

Show: co-subject notification and request flow; sender accept/reject.

21. **What do you think about that idea?**

    Linked: RQ4; RQ5.

    Probes optional: fairness; invasiveness; complexity.

22. **For this kind of scenario where the photo includes the sender and other co-subjects, who do you think should have the strongest control in that situation: the sender, the person in the photo, both equally, or someone else?**

    Linked: RQ4.

23. **How realistic does this feel in a real messaging app?**

    Linked: RQ5.

Moderator note: Use only if time allows or if co-subject control is a session focus.

B.8 Section 7: Wrap-up and final reflections

25. **Would this make you feel more in control when sharing personal photos? Why or why not?**

    Linked: RQ5.

26. **Would it make you trust the platform more, less, or no differently?**

    Linked: RQ5.

27. **What part felt most clear or useful?**

    Linked: RQ3; RQ5.

28. **What part felt most confusing, unrealistic, or uncomfortable?**

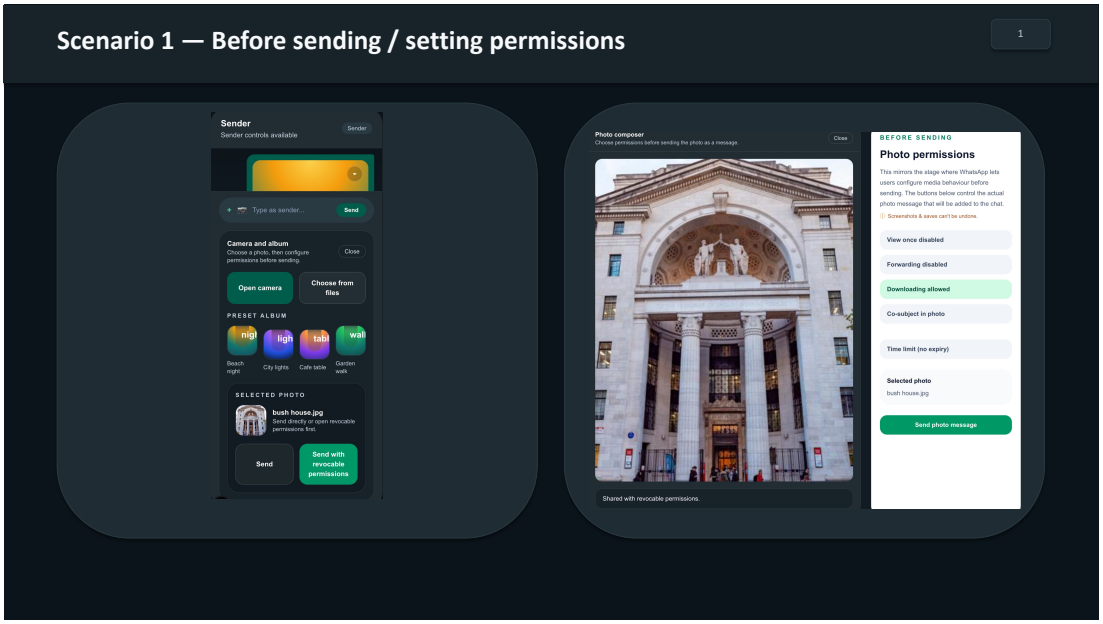
    Linked: RQ1; RQ3; RQ5.

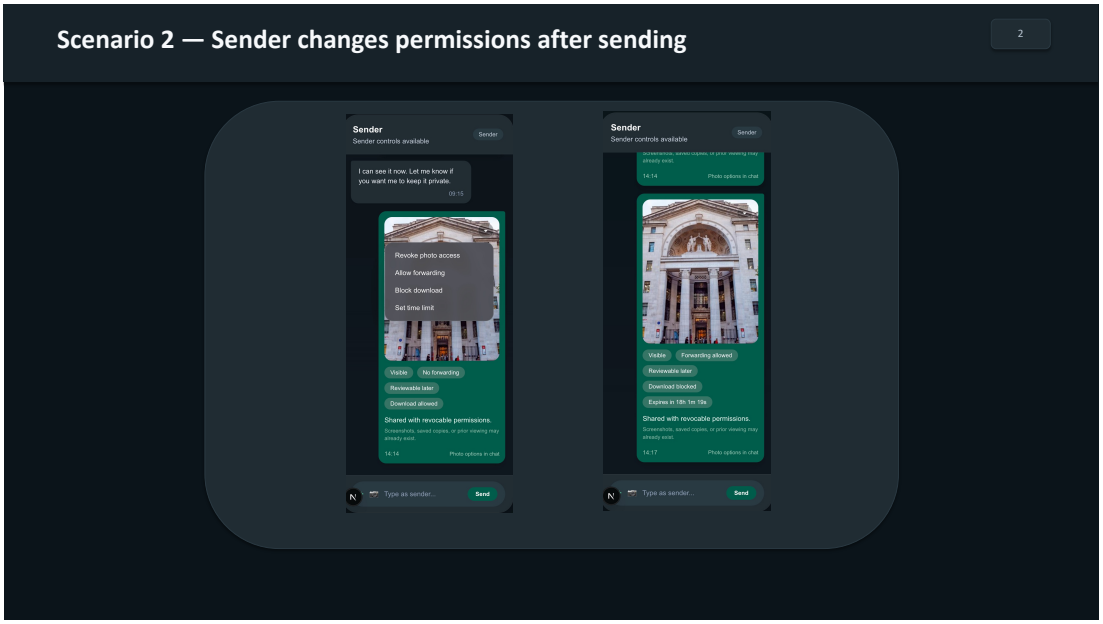
29. **From 1 to 5, how clear do you think everything feels?**

    Linked: RQ3.

B.9 Interview scenario slides (stimulus material)

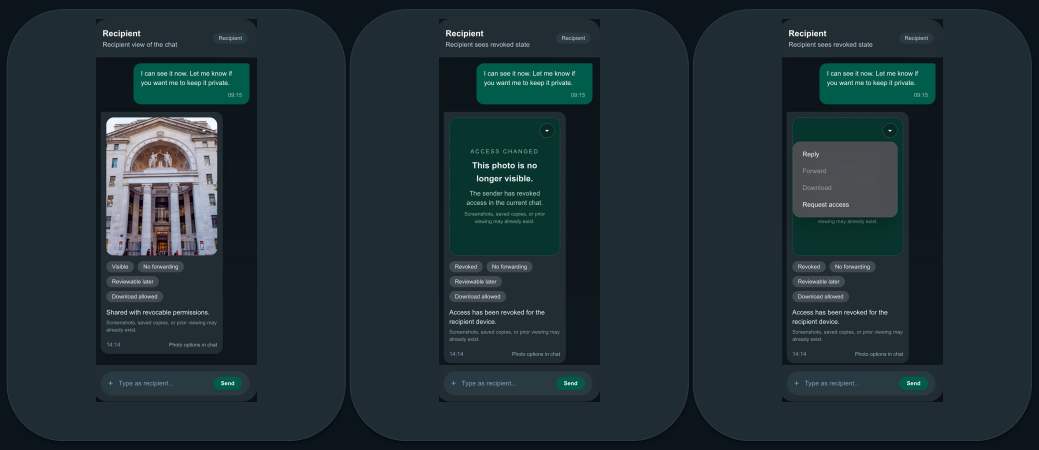
This section reproduces the slide deck *Consent Revocation Interview Scenarios.* The screenshots embedded in these slides were taken from the **first round** of the prototype implementation, before later visual or interaction refinements and they are preserved here to document the materials participants actually saw.





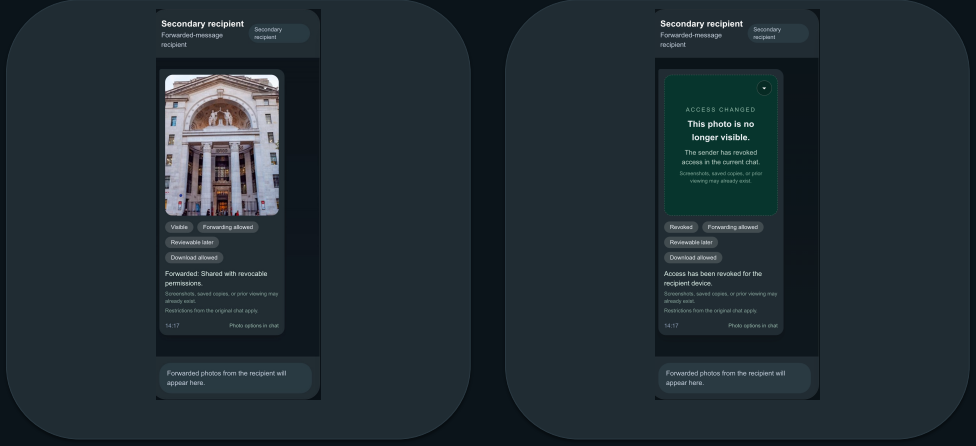
Scenario 3 — Recipient-side impact after access changes

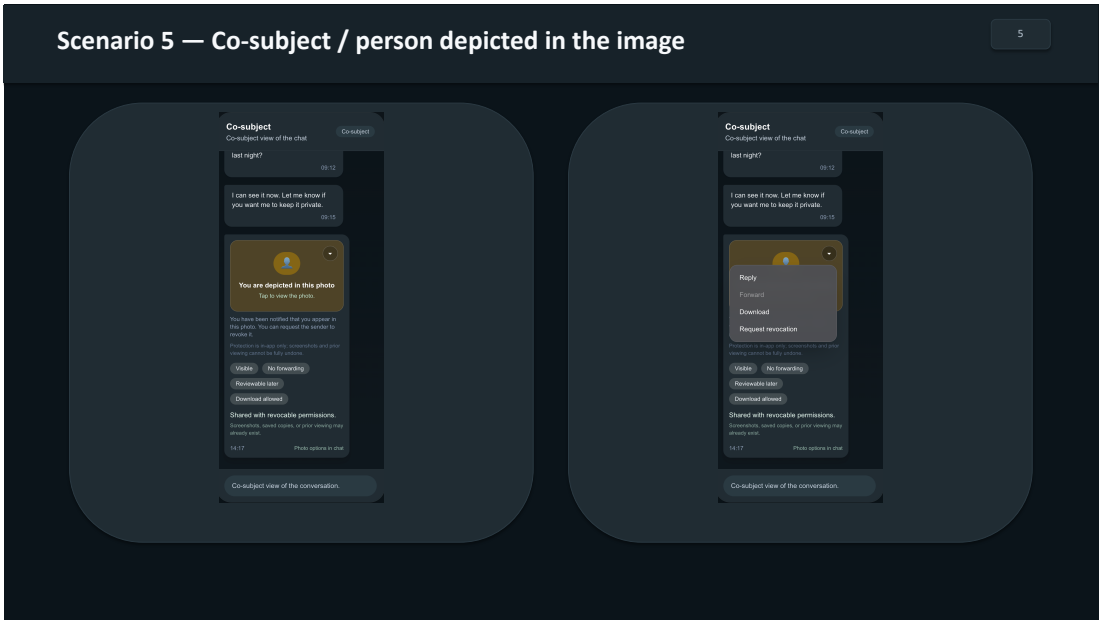
3



Scenario 4 — Forwarding / secondary recipient

4




Scenario 5 — Co-subject / person depicted in the image
5
Co-subject
Co-subject view of the chat
Co-subject
last night?
09:12
I can see it now. Let me know if you want me to keep it private.
09:15
You are depicted in this photo
Tap to view the photo.
You have been notified that you appear in this photo. You can request the sender to revoke it.
Protection is in-app only; screenshots and prior viewing cannot be fully undone.
Visible
No forwarding
Reviewable later
Download allowed
Shared with revocable permissions.
Screenshots, saved copies, or prior viewing may already exist.
14:17
Photo options in chat
Co-subject view of the conversation.
Co-subject
Co-subject view of the chat
Co-subject
last night?
09:12
I can see it now. Let me know if you want me to keep it private.
09:15
Reply
Forward
Download
Request revocation
Protection is in-app only; screenshots and prior viewing cannot be fully undone.
Visible
No forwarding
Reviewable later
Download allowed
Shared with revocable permissions.
Screenshots, saved copies, or prior viewing may already exist.
14:17
Photo options in chat
Co-subject view of the conversation.

Appendix C

User Guide

C.1 Purpose

This appendix explains how to run the **Smart Photos** prototype on a local machine. Also a deployed version can be accessed at `https://smart-photos.vercel.app`. A concise, illustrated tour of every major flow is already given in Section 5.5.

C.2 Prerequisites

- **Node.js** version **20.x** (LTS recommended). The project uses TypeScript tooling aligned with Node 20; older major versions may fail to install or run dependencies.
- **npm** (bundled with Node.js). The instructions below use `npm`

C.3 Install and run

1. Open a terminal and change into the prototype folder:

   ```
   cd photo-revocation-prototype
   ```

2. Install dependencies:

   ```
   npm install
   ```

3. Start the development server:

   ```
   npm run dev
   ```

4. Open the default address printed in the terminal and make it full screen to have a better view.

C.4 Optional: production build

To run an optimised build (closer to a deployed Next.js app, useful for a stable demo):

```
npm run build
npm run start
```

C.5 Feature checklist (where to see each behaviour)

Feature	**What to do in the running prototype**
Pre-send permissions and screenshot warning	From the sender handset, open the photo picker, choose one or more images, then use *Send with revocable permissions*.
Post-send permission changes / revocation	On the sender handset, open the action menu on a sent photo message and use the post-send options (e.g. revoke access, forwarding, download, time limit). Expand the permission summary row under the bubble if needed.
Recipient view and revoked placeholder	Compare the recipient handset before and after sender-side revocation. Use recipient menu actions such as *Forward*, *Download*, or *Request access* when the design allows them.
View-once consumption	On the recipient handset, open a view-once photo via *Open*, then close the viewer and observe the consumed state on the bubble.
Forwarding and access request	Forward from a recipient handset when permitted. After revocation, use *Request access* and respond on the sender handset with *Accept* or *Reject*. Observe the forwarded-recipient panel for propagated state.
Co-subject flow	Use the co-subject handset panel. Read the notification card, open the photo menu, and use *Request revocation* where available. Complete *Accept*/*Reject* on the sender handset.

C.6 Optional: automated tests

```
npm run test:coverage
```

Further detail on testing is given in Section 5.6.

Appendix D

Ethical clearance

This appendix reproduces the ethics approval documentation issued for the user study reported in this dissertation.

The file `appendix-materials/ethics/ethics-approval.pdf` *was not found. Add it to the project (same path as on your local machine) and recompile the appendices document so this page shows the scanned approval.*

Appendix E

Source Code

Authorship statement

I verify that I am the sole author of the programs contained in this folder, except where explicitly stated to the contrary.

Signed: Mert Ayranci **Date:** April 18, 2026

E.1 Submission contents

The complete source code is submitted alongside this report as a separate ZIP archive (named `photo-revocation-prototype.zip`). This appendix is intended as a navigation aid for that archive rather than a verbatim listing. It mirrors the structure of the prototype repository and gives a one-line description of every file or unit.

The prototype is a Next.js (App Router) front-end written in TypeScript, styled with Tailwind CSS, and tested with Vitest and React Testing Library. Build and run instructions are provided in Appendix C; the architectural narrative is given in Chapter 5.4.

E.2 Repository layout

At the top level the archive is organised as follows:

Path	Purpose
`src/app/`	Next.js App Router entry point, root layout, and global stylesheet.
`src/components/photo-revocation-demo/`	React components implementing every interface region of the prototype.
`src/lib/photo-revocation-demo/`	Domain types, helper functions, constants, and initial in-memory state.
`src/lib/`	Cross-cutting permission logic shared between components and tests.
`tests/`	Vitest suite organised into `unit/`, `components/`, `app/`, and shared `helpers/`.
`docs/`	Manual test plan and requirement traceability matrix.
Root configuration	TypeScript, Next.js, Vitest, ESLint, Tailwind PostCSS, and package manifest files.

E.3 File index

The following table lists every file in the submitted archive, grouped by directory, with a one-line description of its responsibility.

Application entry (`src/app/`)

File	Description
`page.tsx`	Thin route component that mounts the `PhotoRevocationDemo` orchestrator on the root URL.
`layout.tsx`	Root layout, metadata, and font configuration applied to every page.
`globals.css`	Tailwind CSS entry point and global style resets.

Domain logic (`src/lib/`)

File	Description
`photoModel.ts`	Pure permission model: `computeEffectiveStatus`, `forwardingAllowed`, `formatExpiryCountdown`, and time-limit helpers.
`photo-revocation-demo/types.ts`	TypeScript types for messages, permissions, participants, and modal state (single source of truth for the data model).
`photo-revocation-demo/constants.ts`	Static configuration: scroll-wheel sizing and the preset photo album.
`photo-revocation-demo/initialState.ts`	Default draft permissions, default time limits, initial conversation, and send-history seeds.
`photo-revocation-demo/helpers.ts`	Derivation helpers used by the UI: `getEffectivePermissions`, `getSourcePolicyPermissions`, view-once consumption checks, and download utilities.

React components (`src/components/photo-revocation-demo/`)

File	Description
`PhotoRevocationDemo.tsx`	Central orchestrator: owns all shared state, handlers, and the four-handset layout.
`PhoneFrame.tsx`	Renders the device chrome for each stakeholder view and provides the per-handset overlay context.
`chatPanelPortalContext.tsx`	React context exposing portal and scroll references so overlays render inside the correct handset.
`MessageList.tsx`	Iterates a message array and dispatches each entry to a text or photo bubble.
`TextMessageBubble.tsx`	Renders text, reply previews, and accept/reject controls for access and revocation requests.
`PhotoMessageBubble.tsx`	Renders photo messages, the collapsible permission summary, role-aware action menu, and revocation submenu.
`PhotoPreview.tsx`	Renders a single photo asset with placeholder, view-once, and revoked variants.
`PhotoStackGrid.tsx`	Multi-photo collage layout for messages with several attached assets.
`PhotoStackLightbox.tsx`	Full-screen gallery viewer that portals into the active handset frame.
`ComposerView.tsx`	Pre-send composer with caption input, permission chips, and the screenshot warning.
`PickerSheet.tsx`	Photo picker sheet with multi-select and the dual *Send*/*Send with revocable permissions* actions.
`ScrollWheelColumn.tsx`	Reusable wheel control used by the time-limit picker.
`TimeLimitModal.tsx`	Post-send modal for adjusting a photo's expiry.
`RequestAccessPanel.tsx`	Inline footer panel for recipients to compose an access request.
`RequestRevocationPanel.tsx`	Inline footer panel for co-subjects to compose a revocation request.
`ViewOnceInChatModal.tsx`	Full-screen viewer that handles the view-once consumption lifecycle.
`ViewOnceSingleTick.tsx`	Small SVG indicator marking a view-once photo as consumed.
`PrototypeStateSummary.tsx`	Side panel exposing the live canonical permission record for evaluator inspection.

Tests (`tests/`)

File names below are listed relative to the `tests/` root and are grouped by sub-directory.

File	Description
`unit/photoModel.test.ts`	Unit tests for the pure permission model and time-limit helpers.
`unit/helpers.test.ts`	Unit tests for derivation helpers, including effective-status fallbacks.
`components/ComposerView.test.tsx`	Component tests for the caption input, permission chips, and the warning banner.
`components/PickerSheet.test.tsx`	Component tests for multi-select and the two send actions.
`components/PhotoMessageBubble.test.tsx`	Branch tests for active, revoked, view-once, and download-disabled variants.
`components/PhotoStackGrid.test.tsx`	Layout tests for multi-photo collages.
`components/PhotoStackLightbox.test.tsx`	Tests for the portalled full-screen gallery.
`components/RequestPanels.test.tsx`	Tests for both inline request panels (access and revocation).
`components/ScrollWheelColumn.test.tsx`	Tests for the duration picker primitive.
`components/TimeLimitModal.test.tsx`	Tests for applying and cancelling post-send expiry changes.
`components/TextMessageBubble.test.tsx`	Tests for replies and request response controls.
`components/ViewOnceInChatModal.test.tsx`	Tests for view-once open, close, and consumption state.
`components/ViewOnceSingleTick.test.tsx`	Snapshot test for the consumed indicator.
`components/PrototypeStateSummary.test.tsx`	Tests that the state panel reflects the canonical record.
`components/PhotoRevocationDemo.test.tsx`	Integration-style journeys covering revocation, forwarding, and request flows across handsets.
`app/page.test.tsx`	Smoke test that the route mounts the orchestrator without error.
`helpers/photoDemoUi.tsx`	Shared Testing Library locators and assertions reused across integration tests.

Documentation (`docs/`)

File	Description
`MANUAL_TEST_PLAN.md`	Manual cross-view verification checklist used after each significant change.
`REQUIREMENT_TRACEABILITY.md`	Mapping from functional and non-functional requirements to components and tests.
`README.md` (root)	Quick-start instructions, architectural overview, and test commands.

Configuration (root)

File	Description
`package.json`	Project manifest, dependency declarations, and `npm` scripts (`dev`, `build`, `start`, `test`, `test:coverage`, `lint`).
`package-lock.json`	Resolved dependency lockfile for reproducible installs.
`tsconfig.json`	TypeScript compiler configuration including the `@/*` path alias.
`next.config.ts`	Next.js framework configuration.
`next-env.d.ts`	Auto-generated Next.js TypeScript ambient declarations.
`vitest.config.ts`	Vitest configuration: jsdom environment, alias resolution, and coverage settings.
`vitest.setup.ts`	Test setup including `@testing-library/jest-dom` matchers.

E.4 Notes for the examiner

- All work on the prototype stopped once the archive was uploaded to KEATS. Several copies of the submitted version are retained locally and one of these copies will be used in the project examination.
- The deployed demonstration at `https://smart-photos.vercel.app` is built from the same submitted source and is provided only as a convenience; the canonical artefact for marking is the submitted ZIP archive.
- No third-party code is included in the archive beyond declared `npm` dependencies, which are installed at build time from `package.json` and `package-lock.json`.